Project Proposal

SignBridge-TR: A Real-Time, Bidirectional AI-Assisted Translation System between Turkish Sign Language, Turkish Text and Speech for Hearing-, Speech- and Visually-Impaired People

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Introduction

Subject area and problem. Our project belongs to assistive technology and draws on computer vision, natural language processing and human–computer interaction. Its topic is automatic translation between a sign language and a spoken language, which has two sides: recognising the signs in a video and turning them into text and speech (sign language recognition and translation), and turning speech or text into signs performed by a virtual character (sign language production).

According to the World Health Organization, about 430 million people (over 5% of the world's population) need rehabilitation for disabling hearing loss, and this number is expected to pass 700 million by 2050 [1]; at least 2.2 billion people have a near or distance vision impairment [2]. In Türkiye, many Deaf people grow up with Turkish Sign Language (TİD) as their first language. TİD has its own vocabulary and grammar; it is not Turkish done with the hands. Law No. 5378 on Persons with Disabilities (2005) gave TİD a legal basis [3], and since 2017 the Ministry of Family and Social Services has published the Current TİD Dictionary online, covering the 2,000 most frequent words [4].

Even so, everyday communication between Deaf and hearing people still relies on interpreters or on notes written on paper. Interpreters are few and usually have to be booked days ahead, and written notes are slow and often fail because many Deaf adults read Turkish as a second language. At a hospital desk, a pharmacy, a bank or a classroom, the hearing person rarely knows TİD. Two groups have an even harder time: speech-impaired people who sign can hear the other person but cannot answer aloud, and blind people can neither see signs nor read written notes, so a blind person and a Deaf person today have almost no way to talk to each other directly.

The problem we want to solve is that there is no tool that TİD users and non-signers, including blind and low-vision people, can pick up at any moment to talk in both directions. This matters because communication barriers turn directly into unequal access to healthcare, education, work and public services, and finding an interpreter for every short conversation is not realistic.

Current research and development. *Sign language recognition:* deep learning on RGB video or on body and hand keypoints is now the norm. There are two large public TİD datasets: AUTSL, with 226 signs, 43 signers and 38,336 isolated samples recorded with RGB, depth and skeleton data [5], and BosphorusSign22k, with 744 signs, 6 signers and 22,542 samples [6]. In the 2021 ChaLearn challenge, multi-modal models that also use skeleton information, such as SAM-SLR, exceeded 98% top-1 accuracy on AUTSL [7]; these results, however, concern

single signs under benchmark conditions and cannot be expected in a live conversation. *Real-time body tracking:* Google's MediaPipe framework tracks 543 landmarks (33 pose, 468 face and 21 per hand) in real time on a phone or in the browser [8], so signs can be recognised on the device without sending video to a server, which protects privacy. *Sign language translation and production:* neural sign language translation was first demonstrated on German Sign Language data [9], and Progressive Transformers showed that skeleton motion can be generated directly from text [10]. In industry, Google announced SignGemma in 2025, which translates American Sign Language (ASL) into English [11]; Hand Talk translates text and speech into Brazilian Sign Language and ASL with 3D avatars [12]; and in Türkiye, Engelsiz Çeviri translates web pages, subtitled videos, PDFs and printed material into TİD using a 3D avatar or a human model [13]. *Speech and language technology:* multilingual speech recognition models such as Whisper transcribe Turkish speech accurately [14], and large language models (LLMs) handle Turkish suffixes and rephrasing well, so we use them to convert between Turkish sentences and TİD gloss sequences (a gloss is the written label of a sign, e.g. [OKUL] for "school").

The gap. While reviewing this work we identified four gaps:

(1) Products in Türkiye work in one direction only: they translate content into TİD. The only product Engelsiz Çeviri lists as "under development" is exactly the missing piece, a two-way tool from speech and text to sign language and back; no method, date or release information has been published [13].

(2) Academic work on TİD mostly stops at recognition accuracy; very few models trained on AUTSL or BosphorusSign22k have been turned into end-to-end conversation systems and tried with Deaf users in a real setting.

(3) Blind users are not taken into account: sign language tools assume the user can see the avatar or read captions, and camera-based tools assume the user can aim the camera. We found no system that lets a blind person and a Deaf person talk to each other.

(4) Models from large companies focus on ASL, whereas TİD has its own vocabulary and grammar (the verb usually comes last, time expressions come first and facial expressions carry meaning) and therefore needs its own data and rules.

Product concept: how SignBridge-TR fills the gap. SignBridge-TR is a web-based progressive web app (PWA) that runs on phones, tablets and computers and translates in real time in both directions (Figure 1):

1. **Direction A – from signs to speech:** camera frames are turned into hand, body and face keypoints on the device; our keypoint-based Transformer recognises the TİD signs; an LLM builds a proper Turkish sentence from the gloss sequence, which is shown in large letters and read aloud.
2. **Direction B – from speech or text to the avatar:** Turkish speech is transcribed; the LLM, together with our TİD rule layer, converts the sentence into a gloss sequence; each gloss matches an animation clip in our sign library, and the 3D avatar plays the clips as a smooth sequence with a caption and adjustable speed. Words not in the library, such as names, are fingerspelled.
3. **Accessibility layer for blind and low-vision users:** every result is read aloud and passed to screen readers (VoiceOver, TalkBack, NVDA); the app can be driven by voice commands; a sound tells the user when the other person's hands are fully in the frame, and a confirmation signals that a message has reached the other side.

A key design decision is that the AI does not generate the avatar's arm and hand movements from scratch: an intermediate layer links glosses to a prepared animation library, so the avatar's signs can be verified and the project stays feasible in two semesters (fully generative motion is future work). In the MVP, recognition works sign by sign, separating signs by the user's short pauses; continuous sign language translation is outside this version's scope. **What we design and implement versus reuse:** we reuse MediaPipe for landmark extraction and existing speech-to-text, text-to-speech and LLM models, but we design and implement the TİD recognition model and its training pipeline, the TİD keypoint dataset and lexicon, the Turkish ↔ TİD rule layer, the animation library and clip-blending engine, the real-time conversation engine (FastAPI + WebSocket) and the accessibility layer.

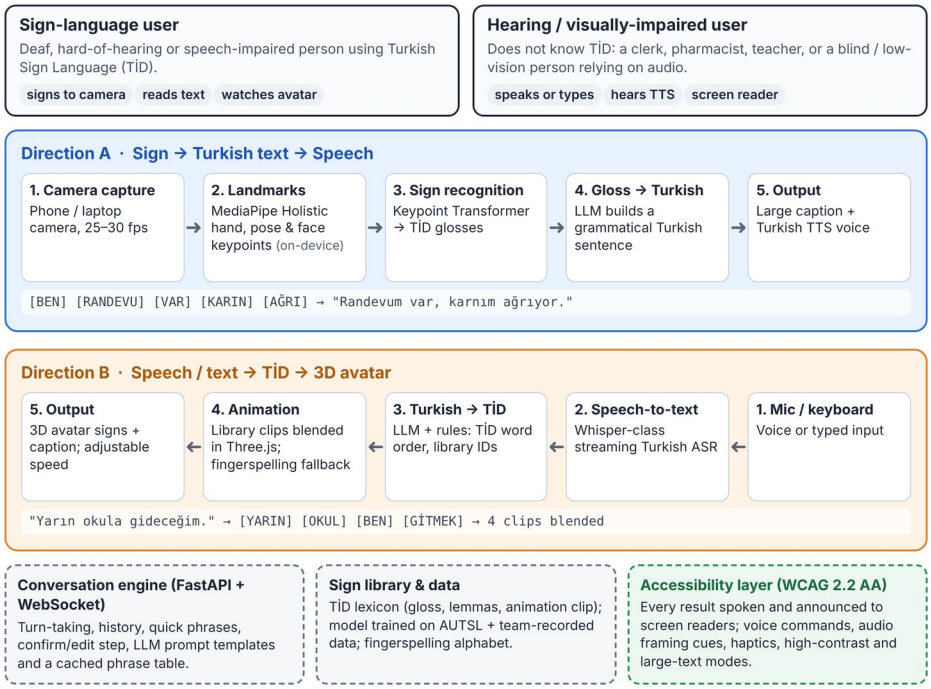


Figure 1 – SignBridge-TR system architecture: two real-time directions share one conversation engine and one accessibility layer; video is processed on the device and only keypoints and text leave it.

Major risks and contingency plans. Table 1 lists the risks that could affect the project and our response to each.

Table 1 – Risk register (L / I = likelihood / impact: H high, M medium, L low)

Risk	L / I	Mitigation and contingency
Not enough TİD training data for the target signs	H / H	Pre-train on AUTSL keypoints; record our own data with at least 10 people and augment it (mirroring, time shifting, keypoint noise); keep the MVP vocabulary to four domains.
The model fails for new people, lighting or backgrounds	M / H	Keypoints are hardly affected by the background; keep test data from people never seen in training; give on-screen and spoken framing guidance; ask the user to confirm a sentence before it is spoken.
Wrong grammar conversion between Turkish and TİD	M / H	Check the rule layer and example prompts against the Current TİD Dictionary (and with a TİD interpreter if one can be reached); limit output to library glosses; fall back to fingerspelling; have 100 test sentences rated by people.
Avatar signs are hard to understand (finger shapes, facial expressions)	M / H	Model the clips on the Current TİD Dictionary and AUTSL videos (or a native signer if available); use an avatar with finger bones and facial expressions; run comprehension tests with Deaf volunteers, otherwise compare clips side by side with the reference videos.
Total delay above the target	M / M	Run the model on the device (ONNX Runtime Web / TF.js, compressed model), stream data over WebSocket, cache frequent phrases, switch to a smaller model if needed.
No TİD interpreter or too few Deaf and blind participants	M / M	Contact the university's disability support office and local Deaf and blind associations early, online if needed; otherwise use the Current TİD Dictionary and AUTSL as sign references, test with fewer participants plus volunteers under simulated conditions (e.g. screen reader only) and report this as a limitation.
Privacy of face and hand video (personal data under KVKK)	L / H	Raw video is never stored or sent; only anonymous keypoints and text leave the device; everyone recorded gives written consent; ethics committee approval is obtained if required.
AI services become paid or unavailable	M / L	Keep open-source alternatives (local Whisper, an open-source LLM, Piper TTS) behind the same service interface; cache responses.

Why we are suitable for this project. Our team of six software engineering students covers the areas the project needs: machine learning with PyTorch, web development with React/Next.js, back-end development with Python and FastAPI, 3D graphics with Three.js and Blender, and user-experience design. We would like language support from a TİD interpreter and to involve Deaf and blind users from the start of the design; because

this support is not yet confirmed, the project does not depend on it (see Table 1). We chose this topic to apply the AI methods we have learned to a problem that directly affects people's daily lives.

Contribution to the UN Sustainable Development Goals. The project contributes most directly to **SDG 10, Reduced Inequalities** [15]: target 10.2 aims for the social and economic inclusion of all irrespective of disability, and we contribute by removing an everyday communication barrier. It also supports three goals indirectly: **SDG 4, Quality Education** (target 4.5: Deaf, speech-impaired and blind students can talk to classmates and teachers without waiting for an interpreter); **SDG 8, Decent Work and Economic Growth** (target 8.5: easier communication at work helps persons with disabilities take part in working life); and **SDG 3, Good Health and Well-being** (target 3.8: access to healthcare becomes easier, and healthcare is where misunderstandings are most dangerous).



UN Sustainable Development Goals [15]; the project contributes directly to SDG 10 and indirectly to SDGs 4, 8 and 3.

Aim and Objectives

Aim. To design, develop and test SignBridge-TR, a real-time, two-way application that translates Turkish Sign Language into Turkish text and speech, and Turkish speech or text into TİD signs performed by a 3D avatar, designed so that Deaf, speech-impaired and blind people can all use it without help from anyone else.

Objectives. Each objective has a measurable success criterion that will be checked at the end of the project (Table 2).

Table 2 – Objectives and success criteria

#	Objective	Success criterion (how it will be measured)
O1	Recognise single TİD signs from a phone or computer camera in real time.	150 signs from health, pharmacy, public services and daily conversation; ≥85% top-1 and ≥95% top-5 accuracy on people never seen in training; ≥15 fps in the browser on a mid-range laptop.
O2	Build grammatical Turkish sentences from recognised gloss sequences.	≥80% of 100 test sequences judged "correct meaning and correct grammar" by two native Turkish speakers; BLEU/chrF scores reported against reference sentences.
O3	Transcribe Turkish speech in streaming mode.	Word error rate ≤15% on a 200-sentence test set recorded in a quiet room.
O4	Translate Turkish into TİD and show it with a 3D avatar.	Animation library of ≥300 signs plus the full Turkish fingerspelling alphabet; ≥75% of signs correctly recognised by Deaf TİD users (if no Deaf participants can be found, ≥75% of signs judged by two independent people to match the Current TİD Dictionary video); ≥70% of 50 test sentences found understandable.
O5	Work fast enough not to break the flow of a conversation.	Median time from the end of a sign to the start of spoken output ≤2.0 s; median time from the end of speech to the avatar starting to sign ≤2.5 s.
O6	Make every flow in the MVP accessible.	Compliance with WCAG 2.2 level AA [16] (automated axe-core scan plus manual checks); all four persona flows completed from start to finish with VoiceOver/TalkBack and with the keyboard only.
O7	Try the system with real users.	≥12 participants: target ≥5 Deaf/hard-of-hearing TİD users, ≥3 blind users and ≥4 hearing non-signers; places that cannot be filled from the target groups are filled with volunteers under simulated conditions and reported as a limitation. ≥80% task completion in the persona scenarios and a mean SUS score ≥70 [17].
O8	Protect users' privacy.	Raw video is never stored or sent (checked by code review and network-traffic inspection); written consent from everyone who is recorded.

User Personas of the Product

Target audience. We expect three main user groups: Deaf and hard-of-hearing people who use TİD; speech-impaired people who hear but communicate in TİD; and blind or low-vision people who need to talk to a TİD user. The hearing people on the other side who do not know TİD are also users: hospital and municipal staff,

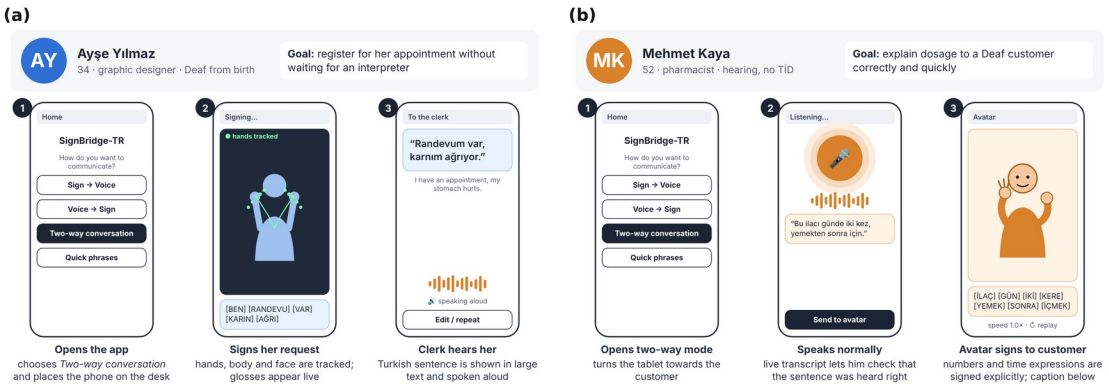
pharmacists, teachers, classmates and family members. On the institutional side, hospitals, pharmacies, municipalities, schools and universities are possible users. Users differ widely in age and comfort with technology, so the interface has to work without any training. From these groups we created four personas (Table 3).

Table 3 – User personas

Persona	Occupation and demographics	Background story	Challenges	What they need to be able to do
Ayşe Yılmaz (Deaf, uses TİD)	Graphic designer at a printing company; 34, female, İzmir. Deaf from birth, TİD is her first language; reads Turkish at a basic level; uses her phone a lot, especially for video calls in TİD.	Went to a school for the Deaf. For hospital appointments she books an interpreter days in advance or takes her sister. When she gets sick suddenly she writes notes, and staff reply by speaking or with long, complicated sentences.	Cannot find an interpreter in emergencies; long written Turkish is hard to follow; staff lose patience; relying on family for health matters takes away her privacy.	Sign comfortably in front of her phone and have it read aloud; see replies in TİD, not only as text, and replay them; start within a few seconds without setup.
Mehmet Kaya (hearing, does not know TİD)	Pharmacist with his own neighbourhood pharmacy; 52, male, Konya. Hearing, does not know TİD; moderate technology skills (uses a tablet for prescriptions).	A few Deaf customers come every month. He explains how to take medicine by writing and gesturing, and worries that a misunderstood “twice a day, after meals” could hurt someone.	Does not know sign language; written instructions are sometimes misread; has to hurry when there is a queue.	Speak normally and have it signed correctly, including numbers and times; understand the customer’s question; use the app without learning anything first.
Zeynep Arslan (blind student)	Third-year psychology student; 21, female, Ankara. Totally blind since childhood; very good with VoiceOver; hearing.	She is in the same project group as her Deaf classmate Emre. They can text to make plans, but face to face they can only talk through a friend who knows a little TİD.	Camera apps assume the camera is aimed correctly; many apps work poorly with screen readers; she cannot tell whether Emre is in frame or whether her message got through.	Use the whole app by voice or with a screen reader; get spoken framing guidance; hear what Emre signs; know that her spoken reply was shown to Emre.
Can Demir (speech-impaired, uses TİD)	Warehouse technician; 28, male, Bursa. Hears well but cannot speak because of a congenital speech disorder; learned TİD at a special-education school.	At a café, shop or bus terminal he types what he wants on his phone and shows it; in a busy queue this is far too slow and people answer before he finishes.	Typing takes time; people do not bother to read notes; text-to-speech apps want typed text instead of signs.	Have what he signs read out at once in a voice he chose; check the sentence before it is spoken; save frequent sentences and reuse them with one tap.

Happy Paths of the User Personas

Figure 2 shows one happy path for each persona; together they cover the core functions of the MVP. The screens are simple wireframe mock-ups and only the successful scenario is shown, from the persona’s goal to the moment the task is completed (labels are in English for readability; the final interface will be in Turkish). (a) Ayşe uses the two-way mode at a hospital desk: her signs are read aloud to the clerk and the clerk’s spoken answer is signed back by the avatar. (b) The flow starts from the hearing side with Mehmet, and numbers and times are signed explicitly. (c) Zeynep completes a similar conversation using only voice and VoiceOver, helped by spoken framing guidance. (d) Can uses only the sign-to-speech mode; he confirms the sentence before it is read out and saves a phrase he uses often.



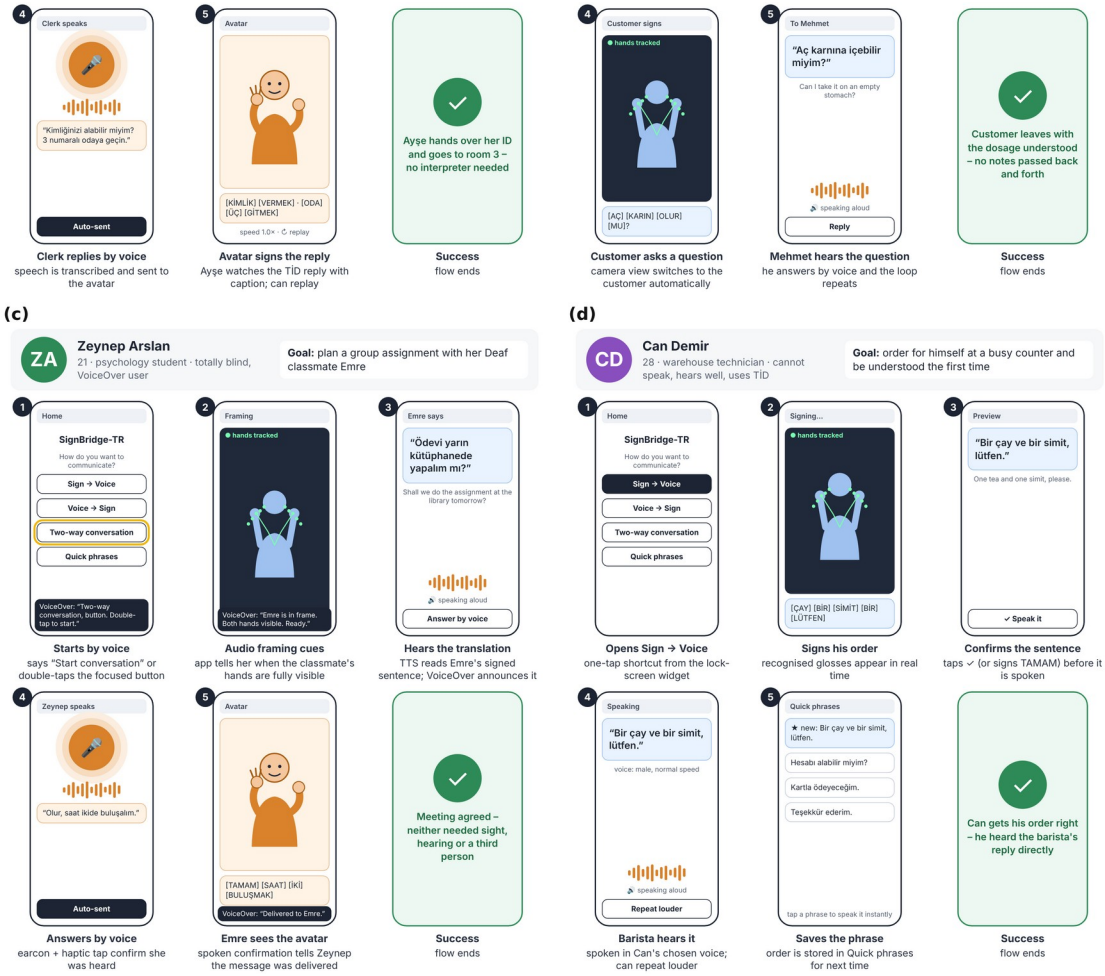


Figure 2 – Happy paths of the four personas: (a) Ayşe at a hospital registration desk, (b) Mehmet explaining a prescription, (c) Zeynep planning a group assignment with her Deaf classmate, (d) Can ordering at a busy café. Steps run left → right, top → bottom.

Expected Outcomes/Deliverables

At the end of the project we will deliver:

1. **SignBridge-TR application:** a PWA with four modes (Sign → Voice, Voice → Sign, Two-way conversation, Quick phrases) that runs in desktop and mobile browsers; source code on GitHub with installation and user documentation.
2. **TİD sign recognition model:** a keypoint-based model exported to run in the browser (ONNX / TF.js), with the training code and an evaluation report (accuracy, confusion matrix, results on people not seen in training).
3. **TİD sign animation library:** at least 300 signs and the fingerspelling alphabet in glTF format; each sign's gloss, Turkish equivalents, domain and clip file kept in a JSON lexicon.
4. **Turkish ↔ TİD translation module:** prompt templates, the rule layer and a 100-sentence Turkish–gloss test set checked against the Current TİD Dictionary (and approved by a TİD interpreter if one joins the project).
5. **TİD keypoint dataset:** at least 150 signs recorded from at least 10 people, stored only as anonymous keypoints and shared publicly if participants agree.
6. **Statement of Work and design documentation:** functional and non-functional requirements, architecture, use-case and UML diagrams, API definitions and interface prototypes.
7. **Test plan and test reports:** unit and integration tests, model comparisons, latency measurements and a WCAG 2.2 AA audit report validating the non-functional requirements.

8. **User study report:** test protocol, task completion rates, SUS results and recommendations for future work, with the ethics committee approval attached if required.
9. **Post-implementation report, demo video, poster and presentation**, together with the sprint records kept in Jira and Confluence.

Related Research and Existing Solutions

Table 4 – Comparison with existing solutions

Solution	Language	Sign → text / speech	Text / speech → sign	Real-time two-way	Blind-user support
Engelsiz Çeviri [13]	TİD	Planned (under development)	Yes (avatar or human model)	Planned (under development)	Not described
Hand Talk [12]	Libras, ASL	No	Yes (3D avatars)	No	Not described
SignGemma [11]	ASL	Yes (to English)	No	No	Not described
AUTSL / BosphorusSign22k research [5]–[7]	TİD	Single signs, offline tests	No	No	No
SignBridge-TR (proposed)	**TİD**	**Yes (text + speech)**	**Yes (3D avatar)**	**Yes**	**Yes (voice, screen reader, spoken guidance)**

Research Questions and Hypotheses

RQ1. How accurately can a keypoint-only model, pre-trained on AUTSL and fine-tuned on our own small dataset, recognise TİD signs from people it has not seen before, in real time, on ordinary devices?

RQ2. Can an LLM supported by a TİD-specific rule layer convert between a Turkish sentence and a TİD gloss sequence without losing the meaning (judged by TİD users or, if they cannot be reached, by raters using the Current TİD Dictionary)?

RQ3. Can Deaf TİD users understand avatar signs built from the library well enough to complete a short service conversation?

RQ4. Can a blind user talk with a Deaf person using only voice and a screen reader?

Hypothesis 1: the keypoint-based Transformer reaches at least 85% top-1 accuracy on the 150-sign vocabulary for people not seen in training. **Hypothesis 2:** adding the rule layer and example prompts significantly increases the share of meaning-preserving conversions compared with the LLM alone. **Hypothesis 3:** participants complete at least 80% of the scenario tasks and the mean SUS score is at least 70.

Methods

Development process. We follow Scrum as defined in the Senior Project Handbook. Work is organised in two-week sprints on Jira with one epic per sub-system (front-end, back-end, ML, avatar, data); stories are estimated with Agile Poker on a 1-2-3-5 point scale in refinement meetings. Each sprint includes daily stand-ups, a sprint review with our advisor acting as Product Owner, a RetroTool retrospective, sprint planning and at least one technical meeting whose action points become Jira issues; all records are kept in our Confluence space. The team-leader role rotates every sprint. Code is integrated on GitHub with GitHub Flow: one branch per Jira issue (story/, task/ or bug/ followed by the issue ID), code review in pull-request discussions, a Review column on the Kanban board, release tags and continuous integration with GitHub Actions. Technical ownership is split into six areas: back-end and LLM integration; computer vision/ML; data and TİD lexicon; 3D avatar and animation; front-end and accessibility; speech, evaluation and testing.

Technical approach.

- **Sign recognition:** MediaPipe Holistic runs in the browser; hand (2 × 21), upper-body and selected face points are normalised for position and scale; a lightweight Transformer encoder classifies 32–64-frame windows, compared with a spatio-temporal graph network. Signs are separated using hand motion and pauses. The model is pre-trained on AUTSL keypoints, fine-tuned on our recordings and exported to ONNX/TF.js.
- **Gloss → Turkish:** an LLM with example prompts answers in a fixed JSON format; frequent sentences come from a cache.

- **Turkish → TİD gloss:** the sentence is analysed morphologically and reduced to word roots (e.g. with Zemberek); the LLM reorders it according to TİD structure (time and topic first, verb last, question sign last); each word is matched to a library sign, then to synonyms, and otherwise fingerspelled.
- **Avatar animation:** a human-like avatar with finger bones and facial expressions; each sign is recorded from a native signer if possible, otherwise from a team member copying the Current TİD Dictionary video; the 3D points are retargeted with inverse kinematics, cleaned in Blender, saved as glTF and blended in Three.js with adjustable speed.
- **Speech:** a Whisper-class Turkish speech-to-text model in streaming mode and a Turkish text-to-speech engine with several voices.
- **Accessibility:** ARIA live regions for screen readers, the Web Speech API for voice commands, framing alerts based on hand-keypoint visibility, and haptic feedback through the Vibration API where supported (Android; on iOS, where web vibration is unavailable, audio cues are used instead).

Data collection, ethics and evaluation. Before any recordings or user tests we will find out whether the university requires ethics committee approval and, if so, apply in the first month. Every participant receives an explanation of the study and gives written consent; personal data are handled under the Turkish Personal Data Protection Law (KVKK, Law No. 6698), and only anonymous keypoints are kept. The evaluation combines numerical measurements (top-1/top-5 accuracy, BLEU/chrF and human ratings, word error rate, latency, axe-core and manual WCAG checks) with a user test based on the four persona scenarios, followed by the System Usability Scale [17] and short semi-structured interviews.

Resource Requirements

Table 5 – Resources

Type	Requirement
Hardware	A GPU for training (the university GPU server if available, otherwise Google Colab, Kaggle or team members' GPUs; keypoint models are light). For recording: two 1080p webcams, a ring light, a tripod and a plain backdrop. For testing: two Android phones, one iPhone and laptops.
Software	Python, PyTorch, MediaPipe, ONNX Runtime Web / TensorFlow.js, FastAPI, React/Next.js, Three.js, Blender, Docker, Jira, Confluence, GitHub and a design tool for prototypes (all free or with student licences).
AI services	API credits for the LLM, speech-to-text and text-to-speech services; otherwise open-source alternatives (local Whisper, an open-source LLM, Piper TTS).
Data	The AUTSL dataset (academic licence), the Current TİD Dictionary as a reference (terms of use to be checked) and our own recordings.
People	Desirable but not yet confirmed: a TİD interpreter as language consultant and Deaf, speech-impaired and blind volunteers via the university's disability support office and local associations; otherwise dictionary/AUTSL references and simulated tests with volunteers.

Project Timeline

Table 6 – Work plan (October 2026 – June 2027)

Work package	Oct	Nov	Dec	Jan	Feb	Mar	Apr	May	Jun
Requirements, literature review, ethics approval process	■	■							
User research, interface prototypes		■	■						
Data recording, lexicon (150 + 300 signs)		■	■	■					
Recognition model v1 → v2			■	■	■				
Avatar rig and animation library			■	■	■	■			
Speech, LLM and TTS pipeline, back-end			■	■	■				
Front-end and accessibility layer				■	■	■			
MVP integration and internal testing						■	■		
WCAG audit and user study							■	■	
Post-implementation report, demo video, poster, presentation								■	■

Use-Case Point Complexity

Following the Guidelines and the Statement of Work template, Table 7 declares the technical complexity factors we consider important for the project; the final decision rests with the reviewers.

Table 7 – Technical complexity factors

Factor	Description	Considered	Factor	Description	Considered
T1	Distributed system	No	T8	Portability to other platforms	Yes
T2	Response time / performance objectives	Yes	T9	System maintenance (deployment)	No
T3	End-user efficiency	Yes	T10	Concurrent / parallel processing	Yes
T4	Internal processing complexity	Yes	T11	Security features	Yes
T5	Code reusability	No	T12	Access for third parties (APIs)	Yes
T6	Easy to install	Yes	T13	End-user training	No
T7	Easy to use	Yes			

Importance and justification. *High:* T2 – median end-to-end latency ≤2.0 s / ≤2.5 s and ≥15 fps on-device inference (O1, O5); T3 – replaces interpreter booking and handwritten notes, with one-tap modes and quick phrases; T4 – keypoint-based sign classification, LLM and rule-based gloss conversion, avatar clip blending; T7 – WCAG 2.2 AA, full screen-reader and voice control, usable without training (O6, O7); T8 – one PWA on Android, iOS and desktop browsers; T11 – on-device video processing, no raw video stored or transferred, KVKK-compliant consent (O8). *Average:* T6 – a PWA opened from a link with no app store or setup; T10 – asynchronous WebSocket streaming of keypoints and audio; T12 – external speech, LLM and TTS services behind one interface with open-source fallbacks.

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